Introduction

WageIndicator Platform Modernization

Technical Proposal for COBRA 2.0 Migration and Unified Architecture Build

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Our Understanding

Strategy

Our Understanding of the Landscape

Analysis of a sophisticated ecosystem spanning 180+ countries and 75+ languages Following an assessment of 14 flow diagrams and 9 analysis documents, we recognize WageIndicator as a vital global entity for labor market transparency. The current ecosystem manages complex datasets for minimum wages, cost of living, living wages, collective bargaining agreements (CBAs), salary benchmarks, labor law, and gig economy indicators using a mix of legacy and modern tools.

Current State

Current Infrastructure Realities

Identifying constraints in the organically grown technology stack

01

COBRA 2003 remains the sole annotation platform for CBA (700 variables) and Labour Law (49 topics) with no failover.

02

30+ ancillary Excel files stored in Dropbox currently power the survey ecosystem via the WageIndicator API.

03

Key-person dependencies exist for Survey XML (Dani/Huub) and Living Wage calculations (Martin Guzi).

04

Manual workflows dominate CBA annotation, Living Wage QA cycles, and legal text updates.

05

Data Governance gaps include a lack of formal versioning, audit trails, and access controls on critical files.

Two Strategic Transformation

Choosing between standard modernization and innovative automation

Version 1: Rebuild

Move to a new foundation while keeping familiar workflows and capabilities.

Version 2: Rebuild + Innovate

Create a smart system with AI annotation, automated pipelines, and QA.

Unified Future

Both paths converge to a scalable, cloud-agnostic architecture replacing legacy limits.

Scope:
Phase 1 & Phase 2

Scope Detail

Phase 1: COBRA 2.0 Migration Scope

Replacing the legacy annotation core with modern workflows

Module	Version 1: Rebuild	Version 2: Innovate
COBRA 2.0 Core Platform	Modern web application: user authentication; role-based access; audit logging. Preserve all existing workflows.	V1 + advanced workflow engine, real-time collaboration, conflict resolution for concurrent users.
Labour Law	49-topic framework with structured templates; annual review workflows.	V1 + automated law change detection via web.
CBA Doc Processing	File upload interface ; basic OCR; format standardization.	V1 + automated doc quality scoring; pipeline for format detection; advanced OCR.
Migration	CBA & Labor Law: PostgreSQL new schema migration and validation.	Identical migration approach for both versions.
CBA Annotation	700-variable coding frame; translation integration.	V1 + AI-powered pre-annotation using LLMs.
Data Export & Integration	REST API and JSON export to Plone CMS. CSV/Excel export.	V1 + REST API option; webhook notifications for data changes; real-time synch.

Scope Detail

Phase 1: COBRA 2.0 Migration Scope contd..

Replacing the legacy annotation core with modern workflows

Module	Version 1: Rebuild	Version 2: Innovate
Translation Management	Centralized translation system; Multi-language interface.	V1 + translation memory with fuzzy matching, automatic translation suggestions via API, legal vocab glossary.
AI Annotation Engine	Not Included.	Complete RAG pipeline; LLM based variable extraction and human-in-the-loop review UI. PoC across 5 languages.

Scope Detail

Phase 2: Unified Platform Build Scope

Modernizing the supporting ecosystem and survey infrastructure

Module	Version 1: Rebuild	Version 2: Innovate
Ancillary Data	Web-based DB replacing 30+ Excel files. Modelled on Minimum Wages Rest API approach.	V1 + automated dependency tracking engine; impact analysis; smart validation rules.
Survey Config	Metadata/Translation UI; XML assist.	V1+ Full drag-and-drop builder; XML auto-gen; eliminates XML dependency
Living Wage Automation	Stata scripts documentation; web-based QA standardized checklist.	Python implementation of Stata calculations; automated QA pipeline; client geo-mapping; historical trend analysis.
Unified Data API	Standardized REST API; basic auth.	V1+ Client SDK generation; OAuth; usage analytics; webhooks.
Monitoring & Observability	Basic health checks and manual log review.	Prometheus + Grafana monitoring dashboards; data freshness tracking; anomaly alerts.

Scope Detail

Phase 2: Unified Platform Build Scope contd..

Modernizing the supporting ecosystem and survey infrastructure

Module	Version 1: Rebuild	Version 2: Innovate
ETL Pipelines	Enhanced Airflow DAGs with monitoring, error notifications, manual retry capability, and audit logging.	V1 + Anomaly detection; real-time monitoring dashboards; self-healing retry logic
KoBoToolbox Integration	Data flow documentation; standardized data import procedures .	Automated db pipeline; field mapping config UI; error handling & reconciliation.
Documentation Platform	Documentation in Markdown/Wiki format; Process runbooks for critical systems.	V1 + API documentation portal (Scalar), onboarding guides, video walkthroughs for key processes.

Scope Boundaries

Out of Scope

Maintaining focus on core infrastructure modernization

01

Plone CMS replacement is excluded; we focus on API compatibility for the current CMS.

02

Public-facing website redesign is out of scope to prioritize backend stability.

03

Salary Survey business logic decisions are separate from technical modernization.

04

Cloud provider selection is a prerequisite workshop; architecture remains cloud-agnostic.

05

Historical data backfilling is separate; migration preserves existing data as-is.

06

KoBoToolbox; Integration architecture is in scope; country-by-country deployment needs separate project management.

07

Living Tariff / Gap Check rebuild; Can be addressed in future phase.

08

Mobile app; Mobile survey apps maintained separately.

09

Full scale AI deployment; V2-AI requirements is limited to RAG and hybrid search.

Critical Path

Assumptions and Dependencies

Critical factors for project success and timeline adherence

01

Data Access: Full access to production COBRA, PostgreSQL, and Dropbox structures is required in Week 1.

02

Domain Experts: Dani, Huub, and Martin Guzi must be available for knowledge transfer.

03

Coding Frames: Existing 700-variable CBA and 49-topic Labour Law structures will be preserved.

04

Infrastructure: Cloud/hosting environments must be provisioned by the WageIndicator IT team by Week 3.

05

AI Sample Data: V2 requires 50+ sample CBA documents across 5 languages for PoC accuracy benchmarks.

06

Documentation Access: V2 requires access to Stata scripts & Living Wage calculation methodology.

07

LLMs: V2 requires decision on LLM provider (Mistral Vs AI Open source).

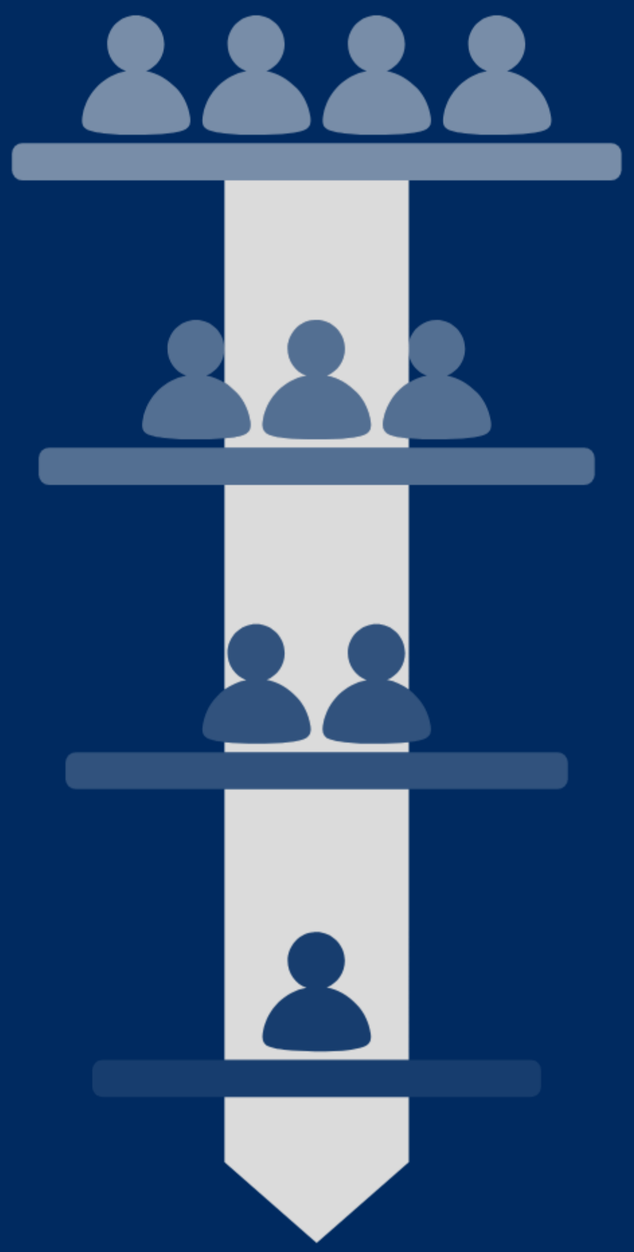
08

Teck-Stack: Open-source technologies preferred. Licensed tools (e.g., LLM API costs) pre-approved by client.

High Level Solution

Architecture Principles

Core tenets guiding the technical design and deployment



API-First

Every component exposes a documented REST API. No direct database access from external systems.

Cloud-Agnostic & Open Source

Runs on any Linux infrastructure (OVH, Hetzner, etc.) without vendor lock-in. All primary components use open-source technologies.

Audit-First

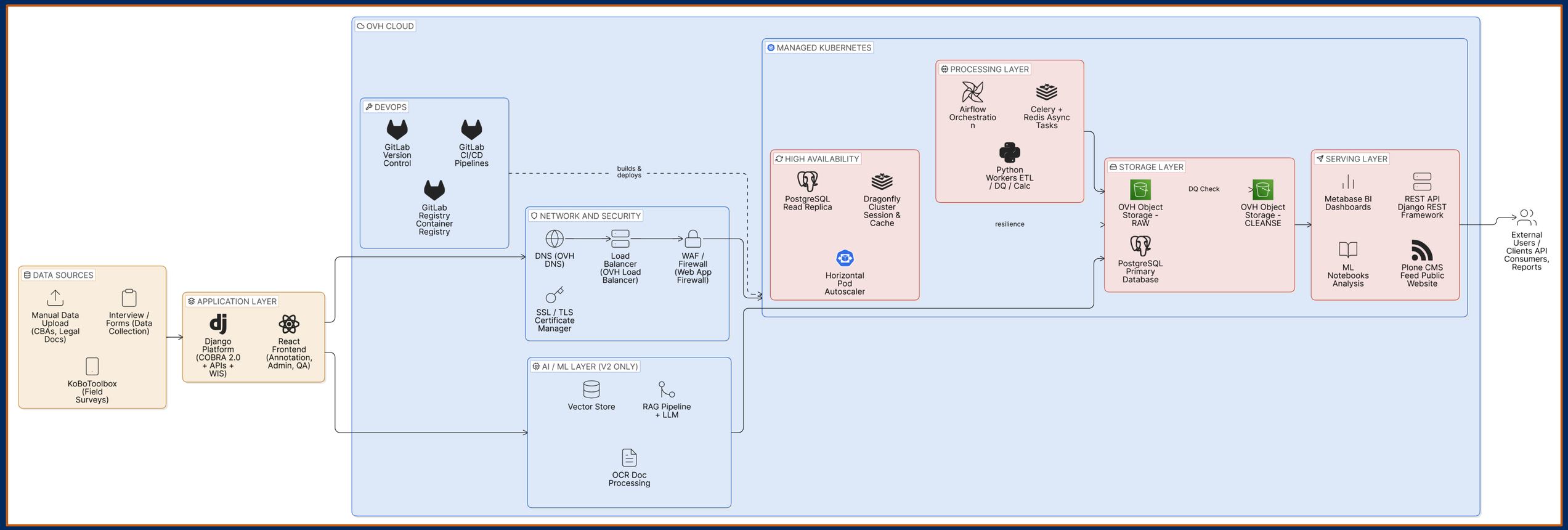
Every data change is logged with timestamp, user ID, and previous values for full accountability.

Backward Compatible

Existing Plone CMS and WageIndicator API consumers & Dropbox outputs continue working throughout the migration.

High-Level Solution Architecture

A modular, API-first ecosystem deployed on OVH Cloud



Both versions (V1 & V2) share the same foundational architecture principles but differ in the intelligence and automation layers

The diagram above illustrates the end-state architecture. All components run on OVH Managed Kubernetes, with OVH Object Storage (S3-compatible) for raw and processed data zones, and Managed PostgreSQL as the primary structured database. The AI/ML Layer (Version 2 only) includes a RAG pipeline for CBA/Labor Law pre-annotation. Gitlab pipelines provides CI/CD.

Tech Stack

Technical Stack Justification

Selected for maturity, community support, and developer availability

Layer	Technology	Rationale
Backend	Python / Django	Existing alignment with Minimum Wages; excellent ORM and auth.
Database	PostgreSQL	WageIndicator standard; ACID compliant; robust JSON support.
ETL Orchestration	Apache Airflow	Already used for COLS ETL; provides DAG-based visualization.
Frontend	React + TypeScript	Component-based architecture for complex data
Infrastructure	Docker / Kubernetes	Ensures consistent deployment on OVH or any cloud provider.
API Framework	Django or FastAPI REST Framework	Native Django integration; auto-generated OpenAI docs.

Tech Stack

Technical Stack Justification contd..

Selected for maturity, community support, and developer availability

Layer	Technology	Rationale
OCR	To Be Decided	Open-source multi-language OCR; no licensing costs.
Doc Processing	To Be Decided	Format detection + text extraction; handles PDF, DOCX, ODF, images
Advanced Search	Elasticsearch	Multi-language analyzers; hybrid search; CBA/legal text search
Advanced Monitoring	Prometheus + Grafana	Real-time dashboards; pipeline health; alerting.
Caching	DragonFly	Session data; API response caching; celery message broker
BI / Reporting	Metabase	Already used for COLS; no-code dashboards

Component Detail

1. COBRA 2.0 Engine Evolution

Upgrading the core annotation technology for better performance. The heart of Phase 1.

Feature	Legacy (2003)	V1: Rebuild	V2: Innovate
Tech Stack	Legacy Web Tags	FastAPI + React	FastAPI + React
Concurrency	Single-user optimized	Multi-user RBAC	V1+ Real-time collaboration; conflict resolution
Search	Basic Lookup	Full-Text (Postgres)	V1+ Hybrid Vector Search
Translation	Separate Windows	Integrated Panel	V1+ Translation Memory; fuzzy matching
Audit Trail	Limited	Full JSON Audit Log	Full JSON Audit Log

Component Detail

1. COBRA 2.0 Engine Evolution contd..

Upgrading the core annotation technology for better performance. The heart of Phase 1.

Feature	Legacy (2003)	V1: Rebuild	V2: Innovate
CBA Workflow	700 vars, single annotator	Same 700 vars, improved UI, multi-user	V1 + AI pre-fills variable suggestions
Labor Law	49 topics, COBRA Edit Form	49 topics, modern editor, templates	V1+ automated law change detection
Doc Processing	Manual upload, external EBBY OCR	Upload + OCR	Automated pipeline: upload, Tika, OCR, HTML conversion
Export	COBRA Dumper (CSV), JSON to Plone	REST API, JSON, CSV, Excel	V1+ REST API, webhooks, analytics
Ai Assistance	None	None	LLM pre-annotation with confidence scoring; AI assistance on Survey

Component Detail

2. AI-Assisted Annotation Pipeline (V2)

Transforming unstructured documents into structured labor data

1

Ingestion & OCR

CBA document upload followed by text and metadata extraction via Apache Tika and Tesseract.

2

Embedding & Indexing

Document chunking and multi-lingual vector embedding for semantic search mapping.

3

LLM Variable Extraction

System retrieves relevant chunks and uses LLMs to extract values for 700 variables.

4

Confidence Scoring

Automatic scoring of extractions; low-confidence items are flagged for mandatory review.

5

Human-in-the-Loop

Annotator reviews suggestions alongside source text to accept, modify, or reject.

Expected impact: 40-60% reduction in annotation time per CBA document, while maintaining or improving accuracy through structured human oversight.

Component Detail

3. Ancillary File Management System

Moving from Dropbox spreadsheets to structured data governance

Web-Based Management

Forms with validation rules (country codes, naming). Eliminates reliance on 30+ disparate Excel files. Direct API integration for downstream consumers

Smart Data Governance

Full role-based access control (RBAC). Versioned snapshots with instant rollback. Audit logs for every single field change

Dependency Engine (V2)

Automated impact analysis for shared files. Notifications for downstream data changes. Smart validation based on historical patterns

Component Detail

4. Living Wage Automation Pipeline

Standardizing complex calculations for accuracy and speed

V1: Process Stabilization

Full documentation of legacy Stata scripts. Web-based QA checklist replacing Google Sheets. Standardized file management for client delivery

V2: Intelligent Automation

Python reimplementation of calculation logic. Automated QA with outlier and trend analysis. Automated client geography mapping (GNSS)

Data Feed

V1: Direct API integration with Numbeo.
V2: Scheduled pulls from the COLS pipeline.
Real-time monitoring via Grafana dashboards

Timelines:
Phase 1 & Phase 2

Phase 1 Timeline: COBRA 2.0 Migration

Version 1 (Rebuild)

3-month focus on like-for-like COBRA replacement

Weeks 1-2

Discovery & Design

Knowledge transfer, system audit, and DB migration plan completion.

Weeks 2-6

Core & CBA Module

Django setup, authentication, 700-variable annotation interface, and OCR integration.

Weeks 6-9

Labour Law Module

49-topic framework, template engine, and COBRA Edit Form replacement.

Weeks 8-11

Integration & API

REST API development, Plone integration, and CSV/Excel export testing.

Weeks 10-12

Testing & Go-Live

Data migration, UAT with domain experts, and production deployment.

Phase 1 Timeline: COBRA 2.0 Migration

Version 2 (Innovate)

5.5-month focus on COBRA replacement + AI Annotation PoC

Weeks 1-3

Discovery & AI Design

Same as V1 plus LLM provider evaluation and AI feasibility assessment.

Weeks 4-10

Platform & CBA / Labor Law Build

Core platform plus advanced workflows and AI integration hooks.

Weeks 10-16

AI Pipeline Development

LangChain setup, RAG pipeline, and variable extraction logic building.

Weeks 14-20

Integration & A/B Testing

REST API development and A/B comparison of Human vs AI annotation.

Weeks 20-22

AI Go-Live

AI model deployment, monitoring, and continuous improvement setup.

Phase 2 Timeline: Unified Platform Build

Version 1 (Rebuild)

4-month focus on modernizing supporting infrastructure

Weeks 1-4

Ancillary File System

DB design for 30+ file types and web form creation for data entry.

Weeks 3-8

Survey Config & Docs

Translation management UI and documentation platform setup.

Weeks 7-10

ETL Enhancement

Airflow DAG development for key pipelines and basic monitoring.

Weeks 9-12

Living Wage Tools

Web-based QA checklist and improved client Excel templates.

Weeks 11-16

API & Deployment

Unified REST API rollout, UAT, and final handover.

Phase 2 Timeline: Unified Platform Build

Version 2 (Innovate)

6-month focus on full platform intelligent automation

Weeks 1-7

Ancillary & Survey Portal

Dependency engine and full drag-and-drop survey builder development.

Weeks 5-12

Living Wage Automation

Python implementation of calculation logic and automated QA pipeline.

Weeks 8-16

Enhanced ETL & API

REST API rollout, webhooks, and data quality gates in Airflow.

Weeks 15-21

KoBo & Monitoring

Real-time KoBo sync and centralized Prometheus/Grafana monitoring.

Weeks 21-24

Full Go-Live

Integrated platform launch, training, and support plan activation.

Team Configuration

Team Config

Team Configuration: Version 1 (Rebuild)

Phase 1 & Phase 2

Core modernization team for a lean rebuilding approach The Version 1 team consists of **4.75 FTE positions** focused on infrastructure stability and like-for-like feature parity. This configuration ensures technical excellence in the core web and database layers.

1 Technical Lead	Architecture, tech decisions, migration planning
1 Sr. Backend Dev	Django, PostgreSQL, API design, data migration
1 Full Stack Dev	React UI, annotation interfaces, unit testing
1 QA Engineer	Test strategy, UAT coordination, migration validation
0.5 DevOps Engineer	CI/CD, Docker, Infrastructure
0.25 Data Engineer	Dataset, Data System, Pipeline, Migration Discussion

Team Config

Team Configuration: Version 2 (Innovate)

Phase 1 & Phase 2

Innovation-ready team with AI and data engineering expertise The Version 2 team expands to **7.0 FTE positions**, adding specialized AI/ML and Data Engineering roles. This team is equipped to handle RAG pipeline development and complex Living Wage automation.

1 Technical Lead	Architecture and AI pipeline design oversight
1 Sr. Backend Dev	Automation, Living Wage logic, and API scaling
1 Full Stack Dev	React UI, annotation interfaces, unit testing
1 Data Engineer	Airflow DAGs, Stata-to-Python, pipeline optimization
1 AI/ML Engineer	LangChain, RAG, LLM extraction, and NLP accuracy
1 QA Engineer	Test strategy, UAT coordination, migration validation, AI accuracy testing
1 DevOps Engineer	CD, Docker, Infrastructure, Prometheus/Grafana

What Does the Finished System Look Like, and What Gets Easier?

The Big Picture

European cloud servers (OVH)

⬇ DATA COMES IN

Staff upload documents • Field workers submit surveys • Forms and data entry

⬇ THE PLATFORM PROCESSES IT

COBRA 2.0 handles annotation • Pipelines check data quality • Calculations run automatically

⬇ SECURITY PROTECTS IT

Encrypted connections • Firewall blocks attacks • Only authorized users get in

IT'S STORED SAFELY	AI ASSISTS (V2)
Secure database Versioned backups Full change history	Reads documents Suggests annotations Saves time

⬇ DATA GOES OUT — UNDER YOUR CONTROL

Your website • Partner reports • Client data feeds • Dashboards

What Gets Easier — Before vs. After

Area	How It Works Today	How It Will Work
CBA Annotation	One person at a time, using a tool built in 2003 with no backup if it fails.	Multiple people work at once, modern interface, full back up. V2: AI does the first pass for you.
Labour Law	Manual tracking of legal changes. Single old editing tool.	Modern editor with templates and version tracking. V2: system watches for law changes automatically.
Key Files	30+ Excel files in Dropbox. No version control. Nobody is sure which files depend on which files.	Online system with version history, validation rules, and rollback. You always know what has changed.
Surveys	Only two people (Dani, Huub) can change the survey configuration. If they're unavailable, work stops.	Survey managers use a simple web tool. V2: drag-and-drop builder, anyone trained can use it.
Living Wage	2-week manual quality check. One person understands the calculation of scripts.	Web-based checklist. V2: fully automated with built-in quality checks.
If Someone Leaves	Critical knowledge walks out the door. No documentation.	Knowledge is built into the system, documented, and any trained person can take over.

Commercials

Thank You

Designing a resilient, scalable, and intelligent foundation to collect, integrate, and analyze global labor data for better policymaking and workforce insights

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